Example 6.4 If θ is a parameter, find the equation of the locus of a moving point, whose coordinates are $(a\sec\theta,\ b\tan\theta)$.

Solution:

Let $P\ (h,\ k)$ be any point on the required path. From the given information we have

$$\begin{aligned} h &= a\sec\theta \text{ and } k = b\tan\theta \\ \Rightarrow \frac{h}{a} &= \sec\theta \text{ and } \frac{k}{b} = \tan\theta \end{aligned}$$

To eliminate the parameter θ, squaring and subtracting, we get

$$\begin{aligned} \left(\frac{h}{a}\right)^2 - \left(\frac{k}{b}\right)^2 &= \sec^2\theta - \tan^2\theta \\ \left(\frac{h}{a}\right)^2 - \left(\frac{k}{b}\right)^2 &= 1 \end{aligned}$$

Therefore the locus of the given point is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

https://www.geogebra.org/geometry

Whenever the parameters are in trigonometric form, try to use trigonometric identities to eliminate θ

$$\boxed{\sin^2\theta + \cos^2\theta = 1,\ \sec^2\theta - \tan^2\theta = 1,\ \operatorname{cosec}^2\theta - \cot^2\theta = 1.}$$

Example 6.5 A straight rod of the length 6 units, slides with its ends A and B always on the x and y axes respectively. If O is the origin, then find the locus of the centroid of ΔOAB .

Solution:

Let the coordinates of the points O, A and B are$(0,0)$, $(a,0)$ and $(0,b)$ respectively.

Observed that the points A and B are moving points.

Let(h,k) be a centroid of ΔOAB.
Centroid of ΔOAB is

$$\left(\frac{0+a+0}{3}\ ,\ \frac{0+0+b}{3}\right) = (h,k).$$

$$\frac{a}{3} = h \ \Rightarrow\ a = 3h, \qquad \frac{b}{3} = k \ \Rightarrow\ b = 3k$$

From right ΔOAB, $\quad OA^2 + OB^2 = AB^2$

$$(3h)^2 + (3k)^2 = (6)^2 \quad \Rightarrow \quad h^2 + k^2 = 4$$

Locus of (h,k) is a circle, $x^2 + y^2 = 4$.

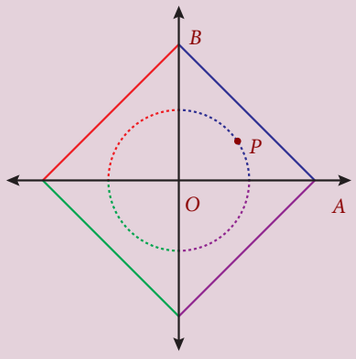


Figure 6.11